

# FIFA 2018 Dataset – ETL Process

## Tools

- Python – data corrections & scraping
- Excel – transformations & validation
- Power Query – data shaping & encoding fixes
- Power BI – data model & visualization

## Extract

- Imported the original FIFA 2018 player dataset.
- Reviewed column types, missing values, and data quality issues.

## Transform

### Data Cleaning

- Fixed UTF-8/special character (e.g. letters with accents) encoding issues present in player names, clubs, and nationalities.
- Removed unnecessary columns (URLs for images of player photos, flags and club logos) since they were outdated and not useful for analysis.
- Corrected FIFA statistics/bonuses (e.g., 95+4 → 99, 98-3 → 95), that caused malformed numeric values.
- Fixed badly parsed numeric fields caused by formatting inconsistencies.

### Monetary Fields

- Cleaned **Value** and **Wage** columns by removing invalid currency characters.
- Converted text values (e.g., “95.5M”, “565K”) into numeric format:
  - K → thousands
  - M → millions
  - B → billions

### Missing Values

- Player attributes (technical/physical stats with range 0-100): replaced missing values with **50** as a neutral/middle value.
- Positional rating columns: replaced missing values with **0** (for goalkeeper positions, since they had no ratings in other stats).

### Feature Engineering

- Created “Main Position” by extracting the first listed position from “Preferred Positions”.

### Validation

- Checked data types after transformations.
- Verified no critical missing values remained.
- Confirmed numeric columns were suitable for calculations and aggregations.

## Load

- Loaded the cleaned dataset into Power BI.
- Created relationships and prepared the data model for dashboard development.

## Visual Asset Preparation

To improve the dashboard visually without using external image URLs, which could slow down Power BI dashboard loading, I prepared local image assets for country flags and club crests. Players' images were not used because of their volume and little analytical value.

### Country Flags

- Extracted the unique nationalities from the dataset.
- Standardized country names when they differed from Wikipedia naming (e.g., St Lucia → Saint Lucia, Korea DPR → North Korea).
- Queried each country's Wikipedia page and extracted the corresponding flag image URL.
- Stored the country list (name from data, search name and URL) in a csv file.
- Downloaded and stored flag images locally for use in Power BI.
- Automatic Success: **163/164** flags –99% (the remaining URL and image was added manually).  
\*Georgia

### Club Crests

- Extracted the unique club names from the dataset.
- Tried direct Wikipedia lookups for each club like the flags.
- Noticed many club names differed from Wikipedia page titles (e.g., FC Barcelona VS Barcelona), implemented a Wikipedia Search API lookup to find the closest matching page (in football) before extracting the crest image.
- Extracted the corresponding crest image URL.
- Stored the club list (name from data, search name after API query and URL) in a csv file.
- Downloaded and stored club crest images locally.
- Automatically retrieved **631/643** crests –98% (the remaining **12** URLs and image were added manually).  
\* AIK, Akhisarspor, Cultural y Deportiva Leonesa, Real Madrid, Sønderjyske Fodbold, Benfica, Vancouver Whitecaps, ADO Den Haag, Club Olimpo de Bahia Blanca, Evkur Yeni Malatyaspor

### HELP

[https://www.mediawiki.org/wiki/API:Parameter\\_information](https://www.mediawiki.org/wiki/API:Parameter_information)

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